Lab 11

Pointing API Manager to a MySQL Database

WSO2 API Manager 4.7.0 · Practical exercise

Overview

Out of the box, API Manager stores its data in embedded H2 databases inside the product folder. H2 is fine for trying the product, but not for production or for a cluster. In this lab you move the two main API Manager databases to MySQL:

Database	deployment.toml section	MySQL script	What it holds
apim_db	`[database.apim_db]`	`dbscripts/apimgt/mysql.sql`	APIs, API Products, applications, subscriptions, OAuth apps and tokens, throttling policies, revisions
shared_db	`[database.shared_db]`	`dbscripts/mysql.sql`	Registry (API definitions, documents, configurations) and the user store (users, roles, claims)

The third section, `[database.local]`, stays on H2. It holds data that belongs to one node only and is never shared.

Note: data is not copied from H2 to MySQL. API Manager starts on MySQL with an empty database (only the admin user and the default policies), so APIs created on H2 in earlier labs do not appear.

Before You Begin

- API Manager 4.7.0 extracted to `<APIM_HOME>`, and **stopped**.
- MySQL 8 server running on port 3306, and the `mysql` command-line client on the API Manager machine.
- The MySQL Connector/J JDBC driver that matches your MySQL version (the Day 3 lab files include `mysql-connector-j-8.2.0.jar` for MySQL 8).
- JDK 21 or 25 set in `JAVA_HOME`, as in the earlier labs.

Placeholders used in this lab:

Placeholder	Meaning
`<APIM_HOME>`	API Manager folder, for example ~/wso2am-4.7.0 or C:\wso2am-4.7.0
`<MYSQL_HOST>`	MySQL host name or IP (localhost when MySQL runs on the same machine)
`<APIM_DB_PASSWORD>`	Password you choose for the apimadmin user
`<SHARED_DB_PASSWORD>`	Password you choose for the sharedadmin user

Part 1 – Create the Databases and Users

1. Log in to MySQL as a user that can create databases (for example root):

macOS / Linux / Windows

```
mysql -h <MYSQL_HOST> -u root -p
```

On Windows, run this from `C:\Program Files\MySQL\MySQL Server 8.x\bin` if mysql is not in the PATH.

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2. Create the two databases. API Manager needs a case-sensitive collation, so do not use the MySQL 8 default (utf8mb4_0900_ai_ci, which is case-insensitive):

```
CREATE DATABASE apim_db CHARACTER SET utf8mb4 COLLATE utf8mb4_bin;
CREATE DATABASE shared_db CHARACTER SET utf8mb4 COLLATE utf8mb4_bin;
```

3. Create one user per database and grant it rights on that database only:

```
CREATE USER 'apimadmin'@'localhost' IDENTIFIED BY '<APIM_DB_PASSWORD>';
GRANT ALL ON apim_db.* TO 'apimadmin'@'localhost';

CREATE USER 'sharedadmin'@'localhost' IDENTIFIED BY '<SHARED_DB_PASSWORD>';
GRANT ALL ON shared_db.* TO 'sharedadmin'@'localhost';

FLUSH PRIVILEGES;
quit;
```

MySQL on another server: API Manager connects from a different host, so create the users for that host instead of 'localhost' (or '%' for any host), and make sure the API Manager machine can reach MySQL on port 3306.

Part 2 – Add the JDBC Driver

4. Copy `mysql-connector-j-8.2.0.jar` to `<APIM_HOME>/repository/components/lib`.

   If API Manager later reports a driver version problem, stop it, remove any other MySQL connector jars from `repository/components/lib` and `repository/components/dropins`, and keep only the one that matches your MySQL version in lib.

Part 3 – Create the Tables

5. Run the WSO2 scripts: the shared script on shared_db as sharedadmin, and the API Manager script on apim_db as apimadmin. Enter the password when asked:

   macOS / Linux

```
mysql -u sharedadmin -p -Dshared_db < '<APIM_HOME>/dbscripts/mysql.sql'
mysql -u apimadmin -p -Dapim_db < '<APIM_HOME>/dbscripts/apimgt/mysql.sql'
```

   Windows (Command Prompt, not PowerShell)

```
mysql -u sharedadmin -p -Dshared_db < "<APIM_HOME>\dbscripts\mysql.sql"
mysql -u apimadmin -p -Dapim_db < "<APIM_HOME>\dbscripts\apimgt\mysql.sql"
```

   Add `-h <MYSQL_HOST>` to each command if MySQL is on another server.

   Important: do not swap the scripts. Running `apimgt/mysql.sql` on shared_db (or the other way round) leaves tables missing, and API Manager fails at start-up.

6. Check that the tables are there:

```
mysql -u apimadmin -p -Dapim_db -e "SELECT COUNT(*) FROM information_schema.TABLES
WHERE TABLE_SCHEMA='apim_db';"
mysql -u sharedadmin -p -Dshared_db -e "SELECT COUNT(*) FROM
information_schema.TABLES WHERE TABLE_SCHEMA='shared_db';"
```

   Expected: apim_db has more than 200 tables (names start with AM_, IDN_ and others) and shared_db has about 50 (names start with REG_ and UM_).

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Part 4 – Configure API Manager

7. Back up `<APIM_HOME>/repository/conf/deployment.toml` (for example copy it to deployment.toml.h2), then open it.

8. Find the two H2 sections at the top of the file, `[database.apim_db]` and `[database.shared_db]`, and replace both with:

deployment.toml

```
[database.shared_db]
type = "mysql"
url = "jdbc:mysql://<MYSQL_HOST>:3306/shared_db?useSSL=false&amp;allowPublicKeyRetrieval=true"
username = "sharedadmin"
password = "<SHARED_DB_PASSWORD>"
driver = "com.mysql.cj.jdbc.Driver"
pool_options.maxActive = 100
pool_options.maxWait = 10000
pool_options.validationInterval = 10000

[database.apim_db]
type = "mysql"
url = "jdbc:mysql://<MYSQL_HOST>:3306/apim_db?useSSL=false&amp;allowPublicKeyRetrieval=true"
username = "apimadmin"
password = "<APIM_DB_PASSWORD>"
driver = "com.mysql.cj.jdbc.Driver"
pool_options.maxActive = 50
pool_options.maxWait = 30000
```

Notes on these values:

- `driver = "com.mysql.cj.jdbc.Driver"` is the driver class for MySQL 8 (Connector/J 8.x).
- MySQL 8 users log in with caching_sha2_password. Without SSL, the driver must be allowed to fetch the server's public key, so `allowPublicKeyRetrieval=true` is required. Without it, API Manager may start the first time but fails after MySQL restarts with `Public Key Retrieval is not allowed`.
- Write `&` between the two URL parameters, not a plain `&`. API Manager copies the URL into an XML file (master-datasources.xml), and a plain & stops the datasources from loading.
- `useSSL=false` keeps the lab simple. In production, enable SSL to MySQL instead.
- The pool options are the sample values from the WSO2 documentation. Size them for your load in production.
- Leave `[database.local]` as it is.

9. Save the file.

Part 5 – Start API Manager and Verify

10. Start API Manager:

macOS / Linux

```
cd <APIM_HOME>/bin
./api-manager.sh
```

Windows (Command Prompt)

```
cd <APIM_HOME>\bin
api-manager.bat
```

Wait for `WSO2 Carbon started in n sec`. There should be no `ERROR` lines about datasources or JDBC. If there are, see Troubleshooting below.

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11. Check that the user store is now in MySQL. The admin user is created in shared_db on first start:

```
mysql -u sharedadmin -p -Dshared_db -e "SELECT UM_USER_NAME FROM UM_USER;"
```

    Expected: two rows, `admin` and `apim_reserved_user` (an internal user that API Manager creates).

12. Log in to the Publisher (`https://<APIM_HOST>:9443/publisher`). The API list is empty, because MySQL started with no APIs. Create a quick test API: **Create API > Start From Scratch**, name `MySQLTest`, context `/mysqltest`, version `1.0.0`, endpoint `https://httpbin.org/anything`, then **Create**.

13. Check that the API is in apim_db:

```
mysql -u apimadmin -p -Dapim_db -e "SELECT API_NAME, API_VERSION, CONTEXT FROM
AM_API;"
```

    Expected: a row for `MySQLTest 1.0.0 /mysqltest/1.0.0`.

14. Restart API Manager. MySQLTest is still in the Publisher, now loaded from MySQL instead of the H2 files in `<APIM_HOME>/repository/database`.

Troubleshooting

Error in wso2carbon.log	Cause and fix
ClassNotFoundException: com.mysql.cj.jdbc.Driver	The JDBC jar is not in <APIM_HOME>/repository/components/lib. Copy it and restart.
Public Key Retrieval is not allowed (Cannot start User Manager Core bundle)	allowPublicKeyRetrieval=true is missing from a URL. Add it to both URLs as in step 8 and restart.
Error in initializing system data sources ... expected a semi-colon after the reference for entity 'allowPublicKeyRetrieval'	A plain & in the URL. Change it to & in both URLs.
Access denied for user 'apimadmin'@'...'	Wrong password, or the user was created for a different host. Check CREATE USER and GRANT in Part 1.
Communications link failure	MySQL is not running, or <MYSQL_HOST> or the port is wrong, or a firewall blocks 3306.
Unknown database 'apim_db'	The database name in the URL does not match the one created in Part 1.
Table 'apim_db.AM_...' doesn't exist	The scripts were not run, or were run on the wrong database. Recreate both databases and repeat Part 3.

Expected Outcome

API Manager runs on MySQL. The API Manager data (APIs, applications, subscriptions, tokens) is in apim_db, and the registry and user store are in shared_db. Data survives restarts, and other API Manager nodes can share the same databases, which is the basis of the active-active deployment lab.

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